JomiCheck

কেনার আগে যাচাই করুন — Check Before You Buy

An AI-powered, lawyer-grade pre-purchase land verification and fraud detection system for citizens of Bangladesh

Event	LandTech 2026 — LandTech Innovation Hackathon
Track	Digital Land Records, Verification & Digitization
Primary problem statement	**P5 — Citizen Pre-Purchase Verification**
Supporting problem statements	P3 — Fraud & Double-Selling Detection (intelligence layer) P2 — Deed & Ownership Verification (tamper-proof report)
Organizer	Dept. of CSE & CSE-TECH, Daffodil International University Supported by the Dept. of Law, DIU
Round 1 deadline	**October 16, 2026**
Document type	Full project plan (internal team document)

Disclaimer: JomiCheck is a decision-support and risk-screening tool. It does not provide official legal verification or legal advice. All prototype data is synthetic. Legal references in this document should be confirmed with the team's faculty mentor / Department of Law before submission.

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1. Executive Summary

Land disputes are among the most common civil cases in Bangladesh. Most of them start before the sale: a seller who is not the recorded owner, land sold twice, an heir selling more than their share, land under litigation, mortgage, acquisition or khas status. An ordinary buyer cannot read and cross-check khatians, deeds, mutation records and legal status on their own.

JomiCheck performs, in minutes, the title search that an experienced land lawyer would carry out before a purchase. A citizen enters the plot identifiers (District, Upazila, Mouza, Dag, Khatian) and the seller's name, or uploads documents. JomiCheck then:

1. Builds the **ownership chain** (বায়া দলিল) and checks it for gaps.
2. Verifies that the seller has the **legal right to sell**: recorded ownership, inheritance share (ফারায়েজ), power of attorney validity, minor/deceased owner.
3. Checks the **status of the land**: litigation, mortgage, khas, vested (অর্পিত), waqf, acquisition, tax arrears.
4. Detects **fraud**: double-selling, over-selling, conflicting registration vs mutation, prior sale agreements (বায়নানামা), suspicious rapid transfers.
5. Warns about **pre-emption risk** (হক-শুফা) from co-sharers.
6. Produces a **transparent risk score** (Low / Medium / High) where every flag cites the relevant law and is explained in simple Bangla or English.
7. Generates a **tamper-proof report** (QR + hash) with a document checklist and next steps.

> **One-line pitch:** JomiCheck does an experienced land lawyer's pre-purchase title search in minutes — ownership, inheritance share, pre-emption, litigation, khas and acquisition — with the relevant law cited for every risk.

2. Hackathon Requirements & Rule Compliance

2.1 Key facts

Item	Requirement
Team	4 members: 2 DIU students (one is Team Leader) + 1 faculty mentor + 1 industry mentor
Round 1 submission	**October 16, 2026** on landtech.diu.edu.bd/landtech
Final Round	Up to 25 teams; Knowledge Tower (Library), DIU; date TBA
Confirmation fee	100 BDT within 3 days of selection announcement (bKash/Rocket)
Final presentation	5 minutes pitch + live demo, 3 minutes Q&A; both students present; mentors cannot answer
Prizes	Champion 20,000 BDT; 1st Runner-up 10,000 BDT; 2nd Runner-up 5,000 BDT

2.2 Strict data, privacy & security rules (Section 8)

> Any violation means **immediate disqualification** and may be reported to authorities.

Rule	How JomiCheck complies
Use only synthetic, organizer-provided or public open data	All owners, plots, deeds, mutations and cases are generated by our own script. Only public administrative names (district / upazila) are real.
No real personal data (names, NID, phone, address, real land records)	Fictional names (e.g. রহিম উদ্দিন), fake NIDs (e.g. `0000000001`), no real addresses.
No scraping / automated querying / bulk download from government portals	No connection to any government system. Integration is described only as future, authorized work.
No security testing of live systems	QR / hash / security features demonstrated only on our own prototype.
Real documents must be anonymized and pre-approved	We use only self-made dummy deeds watermarked "SAMPLE / DUMMY".

2.3 Development & AI rules

- Original work; open-source libraries allowed with licences respected and credited in the README.
- Repository must keep **commit history** — commit early and often, from both students.
- All AI tools used must be **disclosed in the README** with a short note on how they were used.
- Both students must be able to explain all code and design; judges may question either student.
- Changes between Round 1 and Final must be summarized in the README.

3. The Problem in the Bangladesh Context

Land information in Bangladesh is spread across multiple records, offices and survey generations:

Record	What it holds	Office
Khatian / Porcha (খতিয়ান / পর্চা)	Record of rights — owners and their shares	Land Records & Survey / AC Land
Survey generations	CS, SA, RS, BS (and City Survey) khatians — often conflicting	DLRS
Dag (দাগ)	Plot number within a Mouza	Mouza map
Deed (দলিল)	Registered sale, gift (heba), exchange, partition	Sub-Registry Office
Mutation (নামজারি), DCR	Updating the record of rights after transfer	AC Land Office
Land Development Tax (দাখিলা)	Tax payment proof	Union Land Office
Legal status	Court cases, acquisition, khas, vested property, mortgage	Courts, DC office, banks

Common risks a buyer faces

Risk	What happens
Ownership mismatch	The seller is not the person recorded in the latest khatian.
Double-selling	The same plot or share is sold to more than one buyer.
Over-selling	An heir or co-sharer sells more land than their legal share.
Registration vs mutation gap	Deed registered but mutation never done, or done in another name.
Survey conflicts	Owner or area differs between CS, SA, RS and BS records.
Restricted land	Khas, vested, waqf, debottor, acquisition, forest, river-eroded land.
Encumbrance	Land mortgaged to a bank or under an earlier sale agreement.
Pre-emption (হক-শুফা)	A co-sharer later claims the land through court after the sale.
Power of attorney abuse	Sale through a revoked or unregistered POA, or after the principal's death.

The result is lost savings, years of litigation and family conflict. Citizens need a simple, trustworthy way to **screen a plot before paying any money**.

4. Solution Overview & Positioning

4.1 How it works

Step	System action
1	Citizen enters District → Upazila → Mouza → Dag → Khatian and seller's name, or uploads documents.
2	System collects ownership, deeds, mutation, survey history, legal status, mortgage and tax records.
3	OCR / extraction converts uploaded documents into structured fields.
4	Verification engine cross-checks records (ownership, chain of title, survey lineage, mutation).
5	Legal engine checks right to sell (inheritance share, POA, minor/deceased owner, Section 52A).
6	Fraud engine detects double-selling, over-selling, prior bainanama, suspicious transfers.
7	Risk engine computes a tiered, explainable score with legal citations.
8	AI explains findings in Bangla/English; report is generated with QR + hash, checklist and next steps.

4.2 Why JomiCheck stands out

A typical P5 project	JomiCheck
Compares seller name with owner	Full **lawyer-style title search** with a 25-year chain of title
Generic risk score	Tiered score; **every flag cites the relevant law**
Ignores inheritance	**Faraiz share calculator** detects heirs over-selling
Detects double-selling after it happens	**Purchase-intent alert** warns buyers before it happens
No pre-emption awareness	**Hak-e-Shufa risk** warning with co-sharer list
Plain PDF report	**Tamper-proof report** with QR code and SHA-256 hash
Result only	Document checklist, questions for the seller, site-visit checklist, next steps

> Problem statement coverage: **P5** (citizen verification) as the product, **P3** (fraud detection) as the intelligence layer, and **P2** (QR / hashing) for report integrity.

5. Legal Due-Diligence Framework (Lawyer's View)

Before a client buys land, an experienced land lawyer checks the following. JomiCheck automates each step.

#	Lawyer's check	What JomiCheck automates
1	Chain of title (25–30 years)	Builds the deed chain (বায়া দলিল) and flags missing links or a seller who never received title.
2	Survey consistency (CS → SA → RS → BS)	Compares owner and area across survey generations and flags breaks.
3	Mutation, DCR, tax receipts	Verifies mutation is in the seller's name and matches the latest deed; checks tax arrears.
4	Encumbrance	Checks mortgage / lien records and prior registered sale agreements.
5	Restricted status	Checks khas, vested, abandoned, waqf, debottor, acquisition, forest, river-eroded status.
6	Litigation	Checks civil suits, Land Survey Tribunal cases, injunctions, Section 144 proceedings.
7	Seller's right to sell	Section 52A check; inheritance share; POA validity; minor or deceased owner.
8	Co-sharer rights	Pre-emption (Hak-e-Shufa) risk for joint holdings.
9	Physical possession & boundary	Generates a site-visit checklist (possession, চৌহদ্দি, neighbours).
10	Price sanity	Compares declared price with the mouza rate to flag suspicious under-valuation.

6. Complete Feature Set

Priority for Round 1: MUST SHOULD FINAL (deferred to the Final Round)

A. Title & right to sell

Feature	What it does	Priority
Ownership verification	Fuzzy / transliteration-aware match of seller name with the recorded owner and share.	MUST
Section 52A check	Seller's name must be in the latest record of rights, or seller must prove inheritance.	MUST
Ownership history timeline	Visual chain of title with gaps highlighted.	MUST
Faraiz inheritance calculator	Computes each heir's share under Muslim law; shows how much the seller can legally sell.	MUST
Over-selling detection	Flags sale area greater than the seller's legal share.	MUST
Power of attorney check	Registered? Revoked? Principal alive? Scope covers sale?	SHOULD
Minor / deceased owner	Court permission needed for minor's land; all heirs needed for deceased owner's land.	SHOULD
Hindu Dayabhaga inheritance	Inheritance shares for Hindu owners.	FINAL
Survey lineage graph	CS → SA → RS → BS owner/area comparison as a graph.	FINAL

B. Land status & encumbrance

Feature	What it does	Priority
Restricted land check	Khas, vested, abandoned, waqf, debottor, forest, rail/road, river-eroded land.	MUST
Acquisition check	Ongoing or notified government acquisition.	MUST
Litigation check	Civil suits, Land Survey Tribunal cases, injunctions, Section 144.	MUST
Mortgage check	Bank or private mortgage on the plot.	SHOULD
Tax arrears	Unpaid land development tax.	SHOULD

C. Fraud & co-sharer risk

Feature	What it does	Priority
Double-selling detection	Same Dag / share in multiple active transactions.	MUST
Mutation vs registration	Mutation missing, outdated or in a different name than the deed.	MUST
Prior bainanama	Existing registered sale agreement with another buyer.	SHOULD
Hak-e-Shufa warning	Joint holding → lists co-sharers who may claim pre-emption.	SHOULD
Purchase-intent alert	Multiple buyers checking the same Dag are alerted — prevents double-selling.	SHOULD

Suspicious transfer pattern	Rapid repeated transfers or price far below mouza rate (rules + anomaly model).	FINAL

D. Citizen support & report

Feature	What it does	Priority
Risk score with legal citation	Low / Medium / High with reasons and the relevant law for each flag.	MUST
AI explanation (Bangla/English)	Plain-language explanation and follow-up Q&A.	MUST
Verification report (PDF)	Property details, timeline, flags, score, next steps.	MUST
Tamper-proof report	QR code + SHA-256 hash; verify page detects altered copies.	SHOULD
Document checklist	Khatian, DCR, tax receipt, previous deeds, heirship certificate, etc.	SHOULD
Questions for the seller	Generated from the detected risks.	SHOULD
Site-visit checklist	Possession, boundaries, neighbours, access road.	SHOULD
Next-step guide	Which office to visit (AC Land, Sub-Registry, court) and what to bring.	SHOULD
Document upload + OCR	Extract fields from deeds, porcha, mutation documents.	FINAL
Cost calculator	Stamp duty, registration fee, local tax, mutation fee estimate.	FINAL
Map view	Plot location and boundaries (PostGIS).	FINAL

7. Legal Reference Map

> These references guide the rule design. They must be reviewed by the faculty mentor / Department of Law before they are cited in the concept note or the app.

Law	Relevance to JomiCheck
Registration Act, 1908 — Section 52A	Seller's name must be in the latest record of rights (or proof of inheritance) for a sale deed to be registered.
Registration Act, 1908 — Section 17A	Agreement for sale (bainanama) must be registered → prior-agreement check.
Transfer of Property Act, 1882 — Section 53C	Sale of immovable property only by registered deed.
State Acquisition and Tenancy Act, 1950 — Section 96	Pre-emption (Hak-e-Shufa) right of co-sharers within the statutory period.
State Acquisition and Tenancy Act, 1950 — Section 145A	Land Survey Tribunal cases on record-of-rights errors.
Land Crime Prevention and Remedy Act, 2023 (ভূমি অপরাধ প্রতিরোধ ও প্রতিকার আইন, ২০২৩)	Offences such as selling more land than owned, forgery and fraudulent transfer.
Land Development Tax Act, 2023	Land development tax and arrears.
Power of Attorney Act, 2012	Registration, scope and revocation of power of attorney for sale.
Acquisition and Requisition of Immovable Property Act, 2017	Government acquisition status.
Vested Property Return Act, 2001	Vested property (অর্পিত সম্পত্তি) status.
Guardians and Wards Act, 1890	Court permission for sale of a minor's property.
Muslim personal law (Faraiz) / Hindu Dayabhaga law	Inheritance share calculation.

8. Risk Engine & Scoring Model

The engine uses deterministic, explainable rules first. Each rule produces a flag with: severity, points, evidence, legal reference, plain-language explanation and recommended action.

8.1 Severity tiers

Tier	Examples	Effect
CRITICAL	Seller is not the owner; double-selling; khas / vested land; acquisition in progress; revoked POA	Result is **HIGH** regardless of total score
MAJOR	Pending litigation; mortgage; over-selling share; mutation missing or mismatched; survey conflict; prior bainanama	+20 to +30 each
MINOR	Tax arrears; pre-emption exposure; chain-of-title gap; price below mouza rate; document field inconsistency	+5 to +10 each

8.2 Rule weights (initial)

Rule	Tier	Points
Seller ≠ recorded owner	Critical	+40
Same Dag / share in multiple active sales	Critical	+35
Khas / vested / waqf / debottor land	Critical	+40
Acquisition notified or in progress	Critical	+40
Sale through revoked / invalid POA	Critical	+35
Pending litigation / injunction	Major	+30
Sale area exceeds legal share (over-selling)	Major	+25
Mortgage / lien	Major	+25
Mutation missing / in another name	Major	+20
Prior registered bainanama	Major	+20
Survey generation conflict	Major	+20
Conflicting transaction information	Minor	+10
Pre-emption (Hak-e-Shufa) exposure	Minor	+10
Chain-of-title gap	Minor	+10
Price far below mouza rate	Minor	+8
Land tax arrears	Minor	+6

8.3 Score bands

Total is capped at 100. LOW 0–30 MEDIUM 31–60 HIGH 61–100 — any critical flag forces HIGH.

8.4 Flag output format

```
{
  "code": "OWNERSHIP_MISMATCH",
  "tier": "CRITICAL",
  "points": 40,
  "evidence": "Latest BS khatian 456 lists 'Abdul Karim', seller is 'Rahim Uddin'",
  "law": "Registration Act 1908, Section 52A",
  "explain_bn": "বিক্রেতার নাম সর্বশেষ খতিয়ানে নেই...",
  "explain_en": "The seller is not the recorded owner...",
  "action": "Ask the seller for the deed and mutation in his name; verify at AC Land office."
}
```

9. Demo Scenarios

Three prepared cases show the full range of the system. All data is synthetic.

Case 1 — Clean property LOW 6/100

Dag 101, Khatian 210. Seller is the recorded owner, mutation is in his name, complete chain of title, no case or mortgage. Only a small tax arrear.

Recommendation: Low risk. Collect the listed documents, pay the tax arrear and proceed with a lawyer.

Case 2 — Inheritance over-selling MEDIUM 55/100

Dag 118, 1.00 acre. Recorded owner (deceased) left a wife, two sons and one daughter. Seller Karim (a son) is selling 0.40 acre.

Heir	Faraiz share	Area
Wife	1/8	0.125 acre
Son (Karim)	2/5 of remaining 7/8 = 7/20	**0.350 acre**
Son	7/20	0.350 acre
Daughter	7/40	0.175 acre

Flags: over-selling 0.05 acre (+25), mutation not done in heirs' names (+20), Hak-e-Shufa exposure from co-heirs (+10).

Recommendation: Buy only up to 0.35 acre, after mutation, with written consent of the other heirs.

Case 3 — Fraud HIGH 91/100

Seller Rahim, Dag 125, Khatian 456. The latest record shows a different owner, and the same Dag appears in another active transaction.

Flags: ownership mismatch (critical, +40), double-selling (critical, +35), conflicting transaction information (+10), tax arrears (+6).

Recommendation: Do not proceed without official verification at the AC Land and Sub-Registry offices.

10. System Architecture & Tech Stack

10.1 Request flow

Citizen → Web App → FastAPI → Records DB → OCR & Extraction → Verification Engine → Legal Engine → Fraud Engine → Risk Scoring → AI Explanation → Signed Report

10.2 Layers

Presentation	Next.js (React) web app, Bangla + English UI, report viewer, QR verify page
API	FastAPI (Python) — REST endpoints, validation (Pydantic), PDF generation
Verification engine	Ownership match, chain of title, mutation vs registration, survey lineage
Legal engine	Section 52A, Faraiz calculator, POA, minor/deceased owner, Hak-e-Shufa, restricted land
Fraud engine	Double-selling, over-selling, prior bainanama, rapid transfers, price anomaly
Risk engine	Tiered scoring, critical override, legal citation per flag
AI layer	LLM explanation (Bangla/English) and Q&A grounded only on the computed flags
Data	PostgreSQL (SQLite acceptable for Round 1); PostGIS for maps later; synthetic data generator
Integrity	SHA-256 hash of report content, QR code linking to verify endpoint

10.3 Tech stack

Component	Technology
Frontend	Next.js / React, Tailwind CSS
Backend	FastAPI, Pydantic, SQLAlchemy
Database	PostgreSQL (+ PostGIS later); SQLite for quick local setup
Name matching	RapidFuzz + Bangla↔English transliteration table
OCR (Final Round)	PaddleOCR or Tesseract (Bangla + English)
ML (Final Round)	Scikit-learn Isolation Forest for transfer anomalies
AI explanation	Open-source LLM (local) or LLM API; strictly grounded on flags
Report	HTML → PDF, `qrcode` library, SHA-256 (hashlib)
Deployment	Vercel (frontend) + Render / Railway (backend), or a single VPS with Docker

11. Data Model

Entity	Key fields
Location	district, upazila, mouza, jl_no
Plot	dag_no, mouza_id, area_acre, land_class, mouza_rate, geometry (later)
Khatian	khatian_no, survey (CS/SA/RS/BS), mouza_id, plots
KhatianOwner	khatian_id, person_id, share (fraction)
Person	name_bn, name_en, father_name, religion, is_minor, is_deceased, date_of_death, fake_nid
Heir	deceased_id, heir_id, relation (wife, son, daughter, ...)
Deed	deed_no, type (sale/heba/exchange/partition), date, sub_registry, plot_id, area, price, sellers, buyers, status
Bainanama	agreement_no, plot_id, seller, buyer, date, registered, status
Mutation	case_no, khatian_id, new_owner, deed_id, date, status
PowerOfAttorney	poa_no, principal, attorney, scope, registered, revoked, date
LegalStatus	plot_id, type (case/acquisition/khas/vested/waqf/debottor/forest), reference, status
Mortgage	plot_id, lender, amount, date, status
TaxRecord	holding_no, plot_id, paid_upto, arrears
PurchaseIntent	plot_id, buyer_token, created_at
Report	id, inputs, flags, score, level, explanation, sha256, created_at

12. API Design

Endpoint	Purpose
`GET /locations`	District → Upazila → Mouza dropdown data
`POST /verify`	Run full verification for plot + seller; returns flags, score, report id
`GET /plots/{id}/history`	Chain of title and survey lineage
`POST /inheritance/calculate`	Faraiz share calculation for a deceased owner
`POST /explain`	AI explanation / Q&A for a report (Bangla or English)
`GET /reports/{id}`	Report JSON
`GET /reports/{id}/pdf`	Signed PDF report
`GET /verify-report/{hash}`	QR target: confirms whether a report is authentic
`POST /documents`	(Final Round) upload + OCR extraction

13. Synthetic Data Plan

Synthetic data is fictional data created by the team that looks like real land records but belongs to no real person or plot. The hackathon rules require it.

13.1 Dataset size (Round 1)

- 3 mouzas in 2 upazilas (real public district/upazila names; fictional mouza data)
- ~50 plots, ~70 khatians across CS / SA / RS / BS, ~120 fictional persons
- ~150 deeds, ~80 mutations, ~15 legal-status entries, ~8 mortgages, ~6 POAs, ~5 bainanamas

13.2 Planted scenarios (one per rule)

#	Scenario	Expected result
1	Clean plot, full chain, mutation done	LOW
2	Seller not the recorded owner	HIGH (critical)
3	Same Dag in two active sale deeds	HIGH (critical)
4	Heir selling more than Faraiz share	MEDIUM
5	Deed registered, mutation in another name	MEDIUM
6	Khas land offered for sale	HIGH (critical)
7	Plot under acquisition notice	HIGH (critical)
8	Pending civil suit + injunction	MEDIUM / HIGH
9	Bank mortgage on plot	MEDIUM
10	Sale through revoked POA	HIGH (critical)
11	Prior registered bainanama with another buyer	MEDIUM
12	Owner differs between SA and RS khatian	MEDIUM
13	Three transfers within 6 months at low price	MEDIUM

13.3 Rules for the dataset

- Fictional names only (e.g. রহিম উদ্দিন, করিম মিয়া, ফাতেমা বেগম); fake NIDs like `0000000001`.
- Dummy deed images (if any) made by the team and watermarked "SAMPLE / DUMMY".
- Generator script committed to the repo; seed fixed so data is reproducible.
- README statement: *"All data is synthetic and generated by our script. No real personal data or government portal data is used."*

14. AI / ML Layer

Layer	Technique	Round
Rules (core)	Deterministic legal and fraud rules — explainable and auditable	Round 1
Entity matching	Fuzzy matching + Bangla/English transliteration for names	Round 1
Explanation	LLM converts flags into plain Bangla/English; answers follow-up questions; never invents facts — prompt contains only computed flags	Round 1
Document extraction	PaddleOCR / Tesseract + field parser; user can correct extracted fields	Final
Anomaly detection	Isolation Forest on transfer frequency, price/mouza-rate ratio, ownership duration	Final
Workflow (optional)	LangGraph: extract → verify → detect → score → explain	Final

> **Design principle:** the decision comes from transparent rules; AI only explains it. This keeps results trustworthy and defensible in front of law judges.

15. User Interface Screens

#	Screen	Content
1	Home	Tagline, "Start verification" button, language toggle (বাংলা / English)
2	Search form	District → Upazila → Mouza → Dag → Khatian, seller name, area to buy
3	Result dashboard	Risk gauge, level badge, list of flags with tier, law and explanation
4	Ownership timeline	Chain of title with gaps highlighted
5	Inheritance view	Heir tree with Faraiz shares and seller's sellable area
6	Ask AI	Chat about the report in Bangla or English
7	Report	Downloadable PDF with QR, hash, checklists, next steps
8	Verify report	Scan QR / paste hash to check authenticity

16. 3-Day Implementation Plan & Team Split

Student A = backend, data, rules. **Student B** = frontend, report, video, concept note. Both review each other's code daily so both can explain everything.

Day 1 — October 13–14: Foundation

Owner	Task
Both	Create GitHub repo, README skeleton, folder structure; first commits
A	FastAPI project, database models, synthetic data generator with all 13 scenarios
A	Rules: ownership mismatch, double-selling, Section 52A, mutation mismatch, litigation, khas/vested/acquisition
B	Next.js project, search form with cascading dropdowns, result dashboard layout
B	Draft concept note outline

Day 2 — October 15: Differentiating features

Owner	Task
A	Faraiz calculator + over-selling rule; POA, bainanama, mortgage, Hak-e-Shufa, tax rules
A	Risk engine with tiers, critical override and legal citations; `/verify` endpoint complete
B	Ownership timeline, inheritance view, flag cards with law and explanation
B	AI explanation (Bangla/English) using only computed flags
Both	PDF report with QR + SHA-256, document checklist, next steps; verify-report page

Day 3 — October 16: Polish & submit

Owner	Task
Both	Test the 3 demo cases end to end; fix bugs; UI polish
A	Deploy backend + frontend (optional but recommended); finalize README
B	Record and edit 3-minute demo video
B	Finalize 4-page concept note PDF; mentor review of legal references
Both	**Submit before the deadline** — do not wait for the last hour

Cut list if time runs out (in this order)

1. Purchase-intent alert
2. Site-visit checklist and seller questions
3. Verify-report page (keep QR + hash on the PDF)
4. Deployment (submit with a local demo video)

> Never cut: synthetic data with planted fraud cases, core rules, risk score with reasons, Faraiz calculator, Bangla explanation, report.

17. Round 1 Submission Package

17.1 Checklist

✓	Item	Requirement
☐	Concept note PDF	Max 4 pages
☐	GitHub repo link (public or shared with organizers), with commit history	Mandatory
☐	README: setup, data sources, AI tool disclosure, licences/credits	Mandatory
☐	Demo video	Max 3 minutes
☐	Live deployment link	Recommended

17.2 Concept note outline (4 pages)

Page	Content
1	Title, team & mentors, problem statement (P5 + P3, P2), the problem in Bangladesh with key facts
2	Proposed solution, lawyer-grade due-diligence framework, key features
3	Architecture diagram, tech stack, risk engine, synthetic data and compliance
4	Countrywide implementation plan, impact, limitations, future scope

17.3 README must include

- Project overview and problem statement
- Setup and run instructions (backend, frontend, data generator)
- Data sources: "All data is synthetic, generated by `scripts/generate_data.py`"
- AI tools used and how (e.g. coding assistant for scaffolding, LLM for explanations)
- Open-source libraries and licences
- Disclaimer: decision-support tool, not legal advice

17.4 Demo video script (3 minutes)

Time	Content
0:00–0:25	Problem: a buyer loses savings to a seller who sold the same land twice
0:25–0:45	Introduce JomiCheck and what it checks
0:45–1:15	Case 1 — clean plot → LOW
1:15–1:55	Case 2 — inheritance over-selling → MEDIUM, Faraiz view
1:55–2:30	Case 3 — Rahim fraud → HIGH 91/100, flags with laws, Bangla explanation
2:30–2:50	Report PDF with QR; verify page detects a tampered copy
2:50–3:00	Impact and countrywide vision

18. Final Round Plan

18.1 Improvements after Round 1

- OCR document upload with editable extracted fields
- CS → SA → RS → BS survey lineage graph
- Isolation Forest anomaly detection on transfers
- Hindu Dayabhaga inheritance; cost calculator; map view with PostGIS
- Summarize all changes in the README (required)

18.2 Pitch deck (max 10 slides)

#	Slide
1	Title + tagline + team
2	Problem — a real-life style story and key facts
3	Why existing processes fail (fragmented records, no citizen tool)
4	Solution — JomiCheck in one picture
5	Lawyer-grade checks and legal references
6	Live demo (switch to app)
7	Architecture and AI layer
8	Countrywide implementation plan
9	Impact and sustainability
10	Roadmap and ask

18.3 5-minute pitch split

Student A: problem and solution (1:30) → Student B: live demo (2:30) → Student A: architecture, rollout and close (1:00).

18.4 Expected judge questions

Question	Answer direction
Where will real data come from?	Authorized APIs from the Ministry of Land (e-Porcha, e-Mutation, land registration); prototype uses synthetic data as required.
Can forged documents fool the system?	Uploaded documents are cross-checked against official records; in production, the record database is the source of truth, not the seller's papers.
Is the risk score legally valid?	No — it is decision support; every flag cites the law and recommends official verification.
Why not blockchain?	We use hashing + QR for report integrity; a ledger can be added when official records are integrated.

How is the Faraiz calculation done?	Explain the rule-based algorithm (fixed shares, residuaries, 2:1 son/daughter ratio) with an example.
What does the AI do exactly?	Only explains computed flags; it does not make decisions.
Business model?	Free for citizens; paid API for banks, lawyers and real-estate firms; government partnership.

19. Countrywide Implementation Plan

Phase	Timeline	Activities
1. Pilot	0–6 months	Partner with one upazila land office under the Ministry of Land; authorized read-only access; test with real buyers through the office help desk
2. Integration	6–12 months	Official API integration with e-Porcha, e-Mutation, land development tax and land registration systems; court case data where available
3. District rollout	12–24 months	Expand to several districts; Union Digital Centers offer assisted checks for rural citizens
4. National	24+ months	All districts; mobile app and SMS/USSD access; bank and lawyer APIs; official verification QR on reports

Sustainability

- Free basic check for citizens
- Paid API / detailed reports for banks (collateral verification), law firms, developers and real-estate agencies
- Government partnership as part of Smart Land Management / Smart Bangladesh initiatives

Expected impact

- Fewer citizens lose money to fraudulent or risky land purchases
- Fewer land-related civil cases over time
- Better awareness of land law among ordinary buyers
- Pressure on sellers to keep mutation, tax and records up to date

20. Risks & Mitigation

Risk	Mitigation
No access to real government data	Synthetic data for prototype; clear authorized-integration roadmap
Incorrect legal logic	Mentor / Law department review; citations shown; disclaimer
Bangla OCR accuracy	Deferred to Final Round; manual correction of extracted fields
LLM hallucination	LLM receives only computed flags; decisions come from rules
Time pressure (3 days)	Strict MUST / SHOULD / FINAL priorities and cut list
Students unable to explain code	Daily cross-review; each student demos the other's module
Rule violation (data)	Only synthetic data; no portal access; README statement

21. Limitations, Ethics & Disclaimer

- JomiCheck is a **decision-support and risk-screening tool**, not official legal verification and not legal advice.
- The prototype uses only **synthetic data**; it does not access any government system.
- Results depend on the completeness and accuracy of the available records.
- Inheritance calculation covers common cases; complex cases require a lawyer.
- Personal data in a production system must be protected with consent, access control, encryption and audit logs.
- Every report recommends confirming findings at the relevant government office and with a lawyer.

22. Future Scope

- Authorized integration with e-Porcha, e-Mutation, registration and land tax systems
- Survey lineage graph across CS, SA, RS, BS and City Survey
- Map-based plot view and boundary checks with PostGIS and cadastral maps
- Watchlist: "notify me if this Dag is transferred, mortgaged or under a case"
- Mobile app, SMS/USSD access and Union Digital Center assisted mode
- Bank collateral verification API and lawyer dashboard
- Ledger-anchored report hashes when official records are integrated

> **Summary:** JomiCheck turns a lawyer's pre-purchase title search into a fast, transparent and citizen-friendly tool. It fits P5 directly, uses P3 fraud detection as its intelligence layer and P2-style hashing for report integrity, and it is designed to scale countrywide through authorized government integration.